Banking Application

Assignment Report

Kerem Yılmaz

Assignment Description:

In this project, I implemented a basic banking application using the C programming language. The application presents the transactions to be performed to the user through a textual interface and the user can select the transaction to be performed.

The following operations can be performed with the application:

- Creating a new customer account: Defining a new customer account by entering the customer's TR ID Number, name, surname, account number, account balance, currency and other information.
- Account update: Updating the account information of a customer whose account number has been entered.
- Account closure: Closing one or all of a customer's existing accounts.
- Account inquiry: Querying the account of a customer and displaying the details of this account.
- Account listing: Listing of customers who have accounts at the bank.
- Money transfer: Transferring money from one account to another, taking into account exchange rate.
- Creating a receipt: Create a formatted receipt for the account creation, account closing and money transfer transactions performed and save it in a file with the extension of ".txt".

Solution Description:

The application was implemented with the following components:

1. Data Structures and Arrays:

- Arrays were used to store account details such as account number, Turkish ID (TrID), name, surname, currency type, and balance.
- Global variables were declared for ease of data access throughout the program.

2. File Handling:

- Account information and receipts are stored in structured files to ensure data persistence.
- Separate directories were created for accounts and receipts, with subdirectories for different receipt types (e.g., account creation, closure, money transfer).

3. Modular Functions:

- The program consists of modular functions for various operations, such as creating accounts, transferring money, updating account details, and closing accounts.
- Each function is self-contained, enhancing code readability and maintainability.

4. Error Handling and Validation:

- Input validation was implemented for fields such as TrID (11 digits) and balance.
- Error messages are displayed for invalid inputs or operations.

5. Receipts and Logging:

- Detailed receipts are generated for account creation, account closure, and money transfers.
- Each receipt includes information about the transaction and the date and time of occurrence.

Critical Points:

1. Data Validation:

- Ensuring TrID is 11 digits and follows a specific checksum formula.
- Validating balance updates and money transfer operations.

2. File Operations:

- Handling file opening, reading, and writing operations robustly.
- Creating necessary directories dynamically if they do not exist.

3. Currency Conversion:

- Supporting conversions between Turkish Lira (TRY), US Dollar (USD), and Euro (EUR) with predefined exchange rates.

4. Concurrency:

- Avoiding data overwrites or duplication during simultaneous operations

- ## Lessons Learned:

    - **File Handling:** Gained experience in persistent data storage and managing structured file formats.
    - **Input Validation:** Improved skills in validating user inputs and handling edge cases.
    - **Modular Programming:** Learned the importance of modular design for improving code reusability and readability.
    - **C Programming Features:** Developed an understanding of advanced features such as pointers, memory allocation, and time functions for logging transactions.

Example Input and Output:

Here is an example of an input and an output of the code:

```
10 accounts have been loaded successfully.
Welcome to the YilmazBank! What would you like to do?
1) Create Account
2) Money Transfer
3) Account Update
4) Account Inquiry
5) Account Listing
6) Account Closure
7) Exit
2
Please enter the account number you want to transfer money from:
(To exit, type '0')
2
Please enter the amount you want to transfer:
5132
Please enter the account number you want to transfer money to:
5
Money transfer completed.
Do you want to see the receipt?
y/n
y
----------------------------------------
| Date: 2025-01-01                     |
| Time: 23:57:37                       |
|                                      |
|        Money Transfer Receipt        |
|                                      |
| Transaction Summary:                 |
|                                      |
| From:                                |
|                                      |
| Recipient Account Info:              |
|                                      |
| Account Number: 5                    |
| TrID: 4354800                        |
| Name: Ali                            |
| Surname: Bal                         |
| Currency: USD                        |
| Balance Before Transaction:          |
|   USD 3688.00                        |
| Balance After Transaction:           |
|   USD 3831.70                        |
|                                      |
----------------------------------------
The receipt is saved as Banking Application\Receipts\Money Transfer Receipts\213_21_TrID(123)_To_Ali_Bal_TrID(100000000
1)_Money_Transfer_Receipt_2025-01-01_23.57.37.txt to your device, you can check it anytime you want.
Press any key to continue...
```